Chapter 12 — Concurrency & Multithreading for LLD (Java) · Study Edition

> **Study edition.** Every topic of the full chapter, with the padding removed. For complete code listings, full outputs and extra follow-ups, see the Full Edition (`README.md`).
>
> **Why this chapter exists.** In an LLD interview you will hear: *"Two users click **Book** on the same seat at the same moment. What happens?"* A design with no answer is incomplete. This chapter teaches the ideas and Java 21 tools to answer it; Chapter 13 applies them to classic problems, and chapters 14–21 use them in case studies.

Every concept follows: *What it is → The problem it solves → How it works → Java example → Trade-offs → When not to use it → Common mistakes → Real-world examples → In the interview → Self-check → TL;DR.*
Code lives in `code/lld/ch12/<topic>/Main.java`.

```
javac --release 21 -d /tmp/lld_12 $(find . -name '*.java')
java -cp /tmp/lld_12 lld.ch12.threads.Main
```

Part	Sections	Code package
A. Foundations	12.1–12.4	`threads`
B. What goes wrong	12.5 Races · 12.6 Java Memory Model	`race`
C. Locks	12.7–12.11	`locks`
D. Coordination	12.12 Semaphores · 12.13 Conditions · 12.14 Latch & barrier	`coordination`
E. Lock-free	12.15 CAS & atomics · 12.16 Concurrent collections	`atomics`, `collections`
F. Liveness failures	12.17 Deadlock · 12.18 Livelock · 12.19 Starvation	`deadlock`
G. Patterns	12.20–12.24	`patterns`
H. Interviews	12.25 Parking, seat booking, inventory	`lldcases`

Term	Meaning in one line
Thread	One independent sequence of instructions inside a program.
Shared state	Data more than one thread can read or write.
Thread-safe	Correct under any interleaving, with no extra locking by callers.
Atomic	One indivisible step: others see "before" or "after", never "halfway".
Lock / mutex	A token only one thread can hold at a time.
Critical section	Code touching shared state that must not run in two threads at once.
Contention	Many threads wanting the same lock at once.
Blocking	A thread waits (lock, I/O, signal) instead of using the CPU.
Liveness	"Something good eventually happens" — no deadlock, livelock, starvation.
Safety	"Nothing bad ever happens" — no corrupted data, no double booking.

Part A — Foundations

12.1 Concurrency vs parallelism

What it is

Concurrency = *dealing with* many things at once (tasks with overlapping lifetimes, maybe taking turns on one core). **Parallelism** = *doing* many things at the same instant on different cores.

The problem it solves

Programs wait a lot (disk, network, DB, users): concurrency does useful work meanwhile (responsiveness, throughput). Parallelism finishes CPU-heavy work sooner using all cores (speed-up).

How it works

The OS scheduler gives each runnable thread a time slice; when it ends or the thread blocks, a **context switch** saves one thread's registers and loads another's. On one core that is the *illusion* of simultaneity; on many cores threads really run at once.

	Concurrency	Parallelism
Core idea	Structure: many tasks in progress	Execution: many at the same instant
Needs multiple cores?	No	Yes
Main benefit	Overlapping waiting (I/O-bound)	Speed on CPU-bound work
Java tools	Threads, executors, virtual threads, `CompletableFuture`	Fork/Join, parallel streams, `Arrays.parallelSort`
Example	Web server with 10,000 connections	Sorting 100 million numbers

Java example

12.4's `Main` runs 10,000 sleeping tasks on virtual threads in well under a second (concurrency); Chapter 13's merge sort uses Fork/Join (parallelism).

Trade-offs

Concurrency adds complexity and context-switch overhead. **Amdahl's law:** with sequential fraction s, speed-up on N cores $\leq 1 / (s + (1 - s)/N)$ — 10% sequential caps it at 10×.

When not to use it

Small, fast CPU-light tasks (splitting costs more than it saves). Single-threaded designs are valid when they remove all locking (Redis's main thread, Node.js's event loop).

Common mistakes

Mixing up the two words; expecting more threads to always be faster (past the core count, CPU-bound work slows down).

Real-world examples

Concurrency: Tomcat/Jetty thread pools, Netty/Node.js event loops. Parallelism: `Arrays.parallelSort`, parallel streams, Spark/Hadoop.

In the interview

> **30-second answer:** "Concurrency is about structure — several tasks in progress, possibly interleaved on one core. Parallelism is about execution — tasks running at the same instant on several cores. Concurrency helps I/O-bound work; parallelism helps CPU-bound work."

Self-check

- **Q:** 50,000 idle chat connections — concurrency or parallelism? **A:** Concurrency: they mostly wait.

TL;DR

Concurrency = structure; parallelism = simultaneous execution. I/O-bound → concurrency; CPU-bound → parallelism.

12.2 Processes vs threads

What it is

A **process** is a running program with its **own memory**. A **thread** runs *inside* a process, **sharing the heap** but owning its **stack** and program counter.

The problem it solves

Threads share data cheaply within one program; processes give strong isolation (one crash doesn't corrupt others).

How it works

The OS gives a process a private address space and one thread (`main`); more threads share heap objects with separate stacks. Threads communicate through shared objects (with synchronization); processes via pipes, sockets, shared memory, message queues.

	Process	Thread
Memory	Private address space	Shared heap; own stack
Creation cost	High	Lower (platform thread ≈ OS thread, ~1 MB stack reserved); virtual threads much cheaper
Communication	IPC	Shared objects — fast, needs sync
Failure isolation	Strong	Weak (e.g. `OutOfMemoryError` kills all)
Context switch	Heavier	Lighter
Java	`ProcessBuilder`	`Thread`, executors

Java example

```java
// A new PROCESS (separate memory):
Process p = new ProcessBuilder("ls", "-l").inheritIO().start();
p.waitFor();

// A new THREAD (shares this JVM's heap):
Thread t = new Thread(() -> System.out.println("same heap, different stack"));
t.start();
t.join();
```

Trade-offs

Threads: fast sharing, but every shared mutable field is a potential bug. Processes: safe isolation, slower serialized communication.

When not to use it

When isolation matters more than speed (untrusted plugins, crash-prone native code) — use processes or containers.

Common mistakes

Thinking locals are shared (each thread has its own stack); one unbounded platform thread per request (use a pool or virtual threads).

Real-world examples

Chrome tabs as processes; PostgreSQL process-per-connection vs MySQL thread-per-connection.

In the interview

> **30-second answer:** "A process is an isolated program with its own memory; a thread is a path of execution inside a process that shares its heap but has its own stack. Threads are cheaper and share data easily but need synchronization; processes are isolated and safer but communicate through IPC."

Self-check

- **Q:** Two threads increment their own local `int x`. Race? **A:** No — locals live on each stack.

TL;DR

Process = isolated house; thread = worker sharing the heap, owning only its stack. Share cheaply, synchronize carefully.

12.3 Thread lifecycle & states

What it is

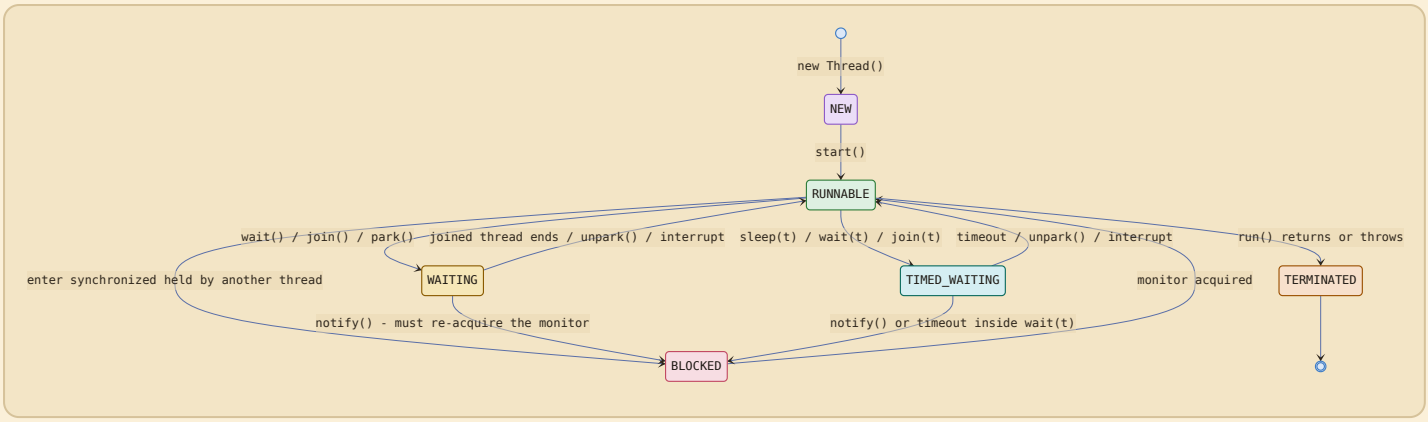
`Thread.getState()` returns one of six `Thread.State` values:

State	Meaning
`NEW`	Created, `start()` not called.
`RUNNABLE`	Running or ready (Java doesn't separate them; most blocking I/O also shows here).
`BLOCKED`	Waiting to *enter* a `synchronized` block held by another thread.
`WAITING`	Waiting indefinitely for another thread: `wait()`, `join()`, `LockSupport.park()` (used by `ReentrantLock`, `Condition`, `CountDownLatch`).
`TIMED_WAITING`	Same with a timeout: `sleep`, `wait(ms)`, `join(ms)`, `parkNanos`, `tryLock(timeout)`.
`TERMINATED`	`run()` returned or threw; cannot be restarted.

The problem it solves

Reading thread dumps (`jstack`, `jcmd <pid> Thread.print`): many `BLOCKED` on one monitor → contention or deadlock; many `WAITING` on a queue → idle pool.

How it works



Calling `start()` twice throws `IllegalThreadStateException`. A thread woken from `wait()` must re-acquire the monitor, so it may pass through `BLOCKED`. **Interruption** is cooperative: `t.interrupt()` sets a flag; `sleep` / `wait` / `join` throw `InterruptedException` and clear it. `Thread.stop()` is unsafe and no longer works.

Java example

`threadStates()` in 12.4's `Main` drives one thread into each state and prints it.

Trade-offs

A model, not a choice — but `RUNNABLE` hides whether a thread is really on a CPU or stuck in native I/O.

When not to use it

Never use `getState()` for program logic (it changes between check and act); monitoring only.

Common mistakes

Calling `run()` instead of `start()` (runs on the current thread); swallowing `InterruptedException` (rethrow or `Thread.currentThread().interrupt()`); confusing `BLOCKED` (monitor) with `WAITING` (signal).

Real-world examples

A `jstack` showing 200 Tomcat workers `BLOCKED` on one `synchronized` logger or pool method = coarse-lock bottleneck.

In the interview

> **30-second answer:** "Java threads have six states: NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, TERMINATED. BLOCKED means waiting for a monitor lock; WAITING/TIMED_WAITING mean waiting for another thread's action, optionally with a timeout."

Self-check

- **Q:** Inside `Thread.sleep(1000)`? **A:** `TIMED_WAITING`.

TL;DR

NEW → RUNNABLE ↔ (BLOCKED | WAITING | TIMED_WAITING) → TERMINATED. BLOCKED = monitor; WAITING = signal. `start()`, never `run()`; respect interrupts.

12.4 Creating threads: `Thread`, `Runnable`, `Callable`, `ExecutorService`, virtual threads

What it is

Way	What you write	Result?	Use when
Extend `Thread`	`class T extends Thread { run() }`	No	Almost never — mixes task and worker
`Runnable`	`() -> { ... }`	No	Fire-and-forget task
`Callable<V>`	`() -> value` (may throw checked)	Yes, via `Future<V>`	Need a result or exception back
`ExecutorService`	`pool.submit(task)`	`Future<V>`	Default: reuse threads, limit concurrency
Virtual threads (Java 21)	`Thread.ofVirtual().start(r)` / `newVirtualThreadPerTaskExecutor()`	via `Future`	Very many *blocking* tasks

The problem it solves

An OS thread per task is expensive and unbounded. Executors separate **what** runs from **how** (pooling, queueing). Virtual threads make "one thread per request" cheap again for blocking code.

How it works

Write a `Runnable`/`Callable`, `submit` it, get a `Future`; `get()` blocks (task exceptions come wrapped in `ExecutionException`); `shutdown()` + `awaitTermination`, or try-with-resources (`close()` waits, Java 19+). **Virtual threads** (JEP 444) run on a small `ForkJoinPool` of carrier threads; blocking I/O, `sleep` or a `java.util.concurrent` lock **unmounts** them. **Pinning** keeps the carrier busy: on Java 21–23 when blocking inside `synchronized` (including `wait()`) or a native method; JDK 24 (JEP 491) removed the `synchronized` case.

Java example

```
// File: code/lld/ch12/threads/Main.java
            Future<Integer> future = pool.submit(sumTask);   // returns immediately
            System.out.println("3) Callable+Future  -> sum 1..100 = " + future.get()); // get() blocks until done
        // ...
        try (ExecutorService vexec = Executors.newVirtualThreadPerTaskExecutor()) {
            List<Future<Integer>> fs = new ArrayList<>();
            for (int i = 0; i < 10_000; i++) {
                fs.add(vexec.submit(() -> { Thread.sleep(50); return 1; }));
            }
            // ... sum the futures
        } // close() waits for all tasks
```

The 10,000 sleeping tasks finish far under 500 s; the file also tours every `Thread.State`. `t.join()` creates a happens-before edge (12.6). Full runnable code: code/lld/ch12/threads/ (see the Full Edition).

Trade-offs

	Platform thread pool	Virtual threads
Cost per thread	High (OS thread + big stack)	Very low (growable heap stack)
How many	Hundreds–few thousand	Millions
Best for	CPU-bound (size ≈ cores)	Blocking I/O
Limits concurrency?	Yes, pool size	No — add a `Semaphore`
Pooling	Yes	Never pool them

When not to use it

Virtual threads for CPU-heavy work (no extra cores); blocking in `synchronized` on Java 21–23 (pinning — prefer `ReentrantLock`); `newCachedThreadPool()` for unbounded traffic.

Common mistakes

Forgetting `shutdown()` (JVM stays alive); ignoring a `Future` (exceptions silently lost); `newFixedThreadPool` for untrusted load — its queue is **unbounded** (use a bounded `ThreadPoolExecutor`, 12.21).

Real-world examples

`Executors` factory methods (Factory pattern); Spring `@Async` on a `ThreadPoolTaskExecutor`; Spring Boot 3.2+ `spring.threads.virtual.enabled=true`.

In the interview

> **30-second answer:** "I never manage raw threads in business code. I describe work as `Runnable` / `Callable` and submit it to an `ExecutorService` sized for the workload: about cores for CPU-bound work, bounded queue with a rejection policy for incoming traffic. For many blocking I/O tasks on Java 21, I use virtual threads — one per task — and protect scarce resources with a semaphore."

Self-check

- **Q:** What does `thread.run()` do? **A:** Runs synchronously on the current thread.

TL;DR

Tasks = `Runnable` / `Callable`; run with an `ExecutorService`; results via `Future`. Virtual threads make one-thread-per-blocking-task cheap. Always shut pools down.

Part B — What goes wrong

12.5 Race conditions & critical sections

What it is

A **race condition**: the result depends on unlucky thread timing. A **critical section**: code touching shared mutable state that must run in one thread at a time. A **data race** (JMM term): two threads access one variable, at least one writes, with no happens-before order.

The problem it solves

Tells you *which* code needs protection and *why*:

Shape	Example	Why it breaks
Read-modify-write	`count++`	Really read, add, write — two threads read the same old value; one update is lost.
Check-then-act	`if (seat.isFree()) seat.book(me);`	The condition can change between check and act.

How it works

1. Find all **shared mutable state**. 2. List its **invariants** ("a seat has at most one owner"). 3. Mark read-then-update code as **critical sections**. 4. Make each **atomic** — a lock, one atomic operation (`incrementAndGet`, `putIfAbsent`), or no sharing (immutability, confinement).

Java example

```
// File: code/lld/ch12/race/Main.java
    /** BROKEN: count++ is read-modify-write (3 steps), so concurrent increments get lost. */
    static class UnsafeCounter implements Counter {
        private int count;
        public void increment() { count++; }
        public int get() { return count; }
    }

    // ... Fix 1: SynchronizedCounter — synchronized increment() AND get() (reads need the lock for visibility)

    /** Fix 2: a lock-free atomic built on compare-and-swap. */
    static class AtomicCounter implements Counter {
        private final AtomicInteger count = new AtomicInteger();
        public void increment() { count.incrementAndGet(); }
        public int get() { return count.get(); }
    }
    // ... check-then-act fixed with one atomic step: seatOwner.compareAndSet(0, id)
```

Output: of 1,000,000 increments the unsafe counter lost e.g. `680044` (varies every run — that unpredictability *is* the race); both fixes give exactly 1000000.

Full runnable code: code/lld/ch12/race/ (see the Full Edition).

Trade-offs

Fix	Pros	Cons
Lock	Protects any multi-step, multi-variable invariant	Waiting; deadlock risk if nested
Atomic variable / map method	Fast, non-blocking	One variable or one map entry only
Immutability / confinement	No synchronization	Needs redesign

When not to use it

Don't lock data that is never shared (locals, confined objects) or never changed after safe construction.

Common mistakes

Protecting writes but not reads; combining two thread-safe calls and assuming atomicity (`containsKey` then `put` on a `ConcurrentHashMap` — use `putIfAbsent`/`computeIfAbsent`); locking on different objects.

Real-world examples

Double-booked tickets, oversold flash-sale stock, double payments on a double-click — check-then-act races, fixed with atomics, locks or DB constraints (unique index, `SELECT ... FOR UPDATE`, version columns).

In the interview

> **30-second answer:** "A race condition is when correctness depends on thread timing. The two common shapes are read-modify-write, like `count++`, and check-then-act, like 'if seat is free then book it'. I find the shared mutable state and its invariants, and make each critical section atomic — with a lock, an atomic operation like `putIfAbsent` or `compareAndSet`, or by removing sharing."

Self-check

- **Q:** Why does `count++` lose updates? **A:** Two threads read the same value before either writes.

TL;DR

Race = result depends on timing. Watch read-modify-write and check-then-act. Make every critical section atomic, including reads.

12.6 Java Memory Model basics: visibility, happens-before, `volatile`

What it is

The **JMM** (JLS ch. 17) defines when one thread's write is guaranteed visible to another's read. **Visibility**: will B see A's write? **Reordering**: compilers, JIT and CPUs reorder when a single thread can't tell. **Happens-before (HB)**: if X HB Y, X's effects are visible to and ordered before Y. `volatile`: reads/writes of that field are visible to all threads and ordered.

The problem it solves

`while (running) iterations++;` with a non-volatile `running` may never stop — the JIT may read it once and loop forever.

How it works

Rule	Meaning
Program order	Within a thread, each action HB the next.
Monitor lock	Unlock HB every later lock of the *same* monitor.
Volatile	A volatile write HB every later read of that field.
Thread start	`t.start()` HB everything in `t`.
Thread join	Everything in `t` HB a return from `t.join()`.
Transitivity	A HB B and B HB C ⇒ A HB C.
Final fields	Separate guarantee (JLS §17.5): after a constructor that doesn't leak `this`, every thread sees correct `final` field values, even without sync.
`java.util.concurrent`	Queue put HB take; `countDown()` HB `await()` return; `Future` completion HB `get()`.

Apply: is every write-then-read pair linked by HB? Same lock → yes. Single independent flag/reference → `volatile`. Compound update → lock or atomic.

Safe publication: share a reference via a `volatile`/`AtomicReference` field, a `final` field, a static initializer, a concurrent collection, or a lock the reader also takes. **Double-checked locking** needs `volatile`:

```java
class Config {
    private static volatile Config instance;          // volatile is essential
    static Config get() {
        Config c = instance;
        if (c == null) {
            synchronized (Config.class) {
                c = instance;
                if (c == null) instance = c = new Config();
            }
        }
        return c;
    }
}
```

(Simpler: the **holder idiom** — `private static class Holder { static final Config I = new Config(); }`.)

Java example

`Worker` in 12.5 uses `private volatile boolean running`; the output `volatile flag stopped worker: true` confirms the write became visible.

Trade-offs

Tool	Visibility	Atomic compound ops	Blocking	Cost
`volatile`	Yes	No	No	Low
`synchronized` / `Lock`	Yes (same lock)	Yes	Yes	Medium
`Atomic*`	Yes	One variable	No	Low–medium
`final`	Yes, after construction	N/A	No	Free

When not to use it

`volatile` for anything combining several steps or variables.

Common mistakes

"Works on my laptop" — x86 reorders less than ARM (Apple Silicon, Graviton); leaking `this` from a constructor; synchronizing only the writer.

Real-world examples

`volatile` shutdown flags; `ConcurrentHashMap`, `AtomicInteger`, `CopyOnWriteArrayList` use volatile + CAS internally; immutable `String`.

In the interview

30-second answer: "The Java Memory Model says a thread is only guaranteed to see another thread's write if there's a happens-before edge — unlock-then-lock of the same monitor, a volatile write-then-read, thread start/join, or a concurrent-collection handoff. `volatile` gives visibility and ordering for a single variable but not atomicity; for compound actions I use locks or atomics."

Self-check

- **Q:** Non-volatile `ready = true` after preparing data; B spins on it. Two bugs? **A:** B may never see it, or see it with stale data.

TL;DR

Cross-thread visibility only via happens-before. `volatile` = visibility + ordering, not atomicity. Locks = both. Publish objects safely.

Part C — Locks

12.7 Mutex (mutual exclusion) and `synchronized`

What it is

A **mutex** is a lock at most one thread can hold. Every Java object has a built-in one — its **intrinsic lock** or **monitor** — taken by `synchronized` (instance method → `this`; static method → the `Class` object; block → the given object).

The problem it solves

Makes a critical section **atomic** and **visible** (the next lock holder sees everything done before the unlock).

How it works

Free monitor → take it; held → `BLOCKED` until free; released automatically on exit, **even by exception**; **reentrant** (the holder can re-enter).

Java example

`SynchronizedCounter` (12.5). Prefer a **private final** lock object so outside code can't take your lock:

```
private final Object lock = new Object();
void add(Item i) { synchronized (lock) { items.add(i); } }
```

Trade-offs

Pros	Cons
Simplest; auto-release; reentrant; cheap when uncontended	No timeout, `tryLock` or interruptible wait; one wait-set per object; no fairness; pins virtual threads on Java 21–23

When not to use it

When you need a timeout, interruptibility, fairness, several conditions, or read/write separation — use `ReentrantLock` / `ReentrantReadWriteLock`.

Common mistakes

Synchronizing on a *changing* reference; on string literals or boxed values (shared JVM-wide); doing slow I/O while holding the lock.

Real-world examples

`Vector`, `Hashtable`, `StringBuffer`, `Collections.synchronizedList` — safe per call but slow under contention, hence replaced by unsynchronized classes plus concurrent collections.

In the interview

> **30-second answer:** "A mutex lets one thread at a time into a critical section. In Java, `synchronized` uses the object's monitor: it's reentrant, released automatically, and gives visibility. I lock on a private final object and keep the section short."

Self-check

- **Q:** Synchronized methods on *different* objects block each other? **A:** No — different monitors.

TL;DR

One holder at a time; `synchronized` is reentrant and auto-releasing. Private final lock object; short sections.

12.8 `ReentrantLock` and reentrancy

What it is

`ReentrantLock` is an explicit mutex with `synchronized`'s meaning plus extras. **Reentrant**: the holder can acquire it again (hold count rises) and must release it as many times.

The problem it solves

Without reentrancy, a locked method calling another locked method on the same object would **deadlock with itself**. Extras over `synchronized`: `tryLock()` / `tryLock(timeout)`, `lockInterruptibly()`, optional **fairness**, multiple `Condition`s, `isLocked()` / `getHoldCount()`.

How it works

`lock()` takes it (count 0 → 1) or waits; the same thread locking again → count 2; each `unlock()` decrements, releasing at 0. **Always** `lock(); try { ... } finally { unlock(); }`.

Java example

```java
// File: code/lld/ch12/locks/Main.java
        void withdraw(int amount) {
            lock.lock();
            try {
                if (canWithdraw(amount)) balance -= amount;   // calls another locked method
            } finally {
                lock.unlock();                                // ALWAYS unlock in finally
            }
        }
        boolean canWithdraw(int amount) {
            lock.lock();                                      // same thread re-acquires: no self-deadlock
            try {
                System.out.println("hold count inside nested call = " + lock.getHoldCount());
                return balance >= amount;
            } finally {
                lock.unlock();
            }
        }
```

Output: `hold count inside nested call = 2`, `balance = 70`. The same file also demos `tryLock` (12.9), `ReadWriteLock` and `StampedLock` (12.10). Full runnable code: code/lld/ch12/locks/ (see the Full Edition).

Trade-offs

Feature	`synchronized`	`ReentrantLock`
Release	Automatic	Manual (`finally`)
Try / timeout	No	`tryLock()`, `tryLock(t, unit)`
Interruptible wait	No	`lockInterruptibly()`
Fairness	No	Optional (slower)
Conditions	One wait set	Many
Reentrant	Yes	Yes

When not to use it

If you need none of the extras, `synchronized` is shorter and can't be left locked.

Common mistakes

`lock()` *inside* `try` (a failed lock would `unlock()` → `IllegalMonitorStateException`); missing `unlock()` on an early return; fairness "to be safe" (much slower).

Real-world examples

`ArrayBlockingQueue`, `LinkedBlockingQueue`, `ThreadPoolExecutor` use it; `ReentrantLock`, `ReentrantReadWriteLock`, `Semaphore`, `CountDownLatch` are built on **AQS**; `CyclicBarrier` on `ReentrantLock` + `Condition`.

In the interview

> **30-second answer:** "`ReentrantLock` is an explicit mutex with the same semantics as `synchronized` plus try-lock with timeout, interruptible acquisition, optional fairness, and multiple conditions. Reentrant means the owner can re-acquire it; the hold count must drop to zero to release. I always unlock in `finally`."

Self-check

- **Q:** `lock()` ×3, `unlock()` ×2 — free? **A:** No; hold count 1.

TL;DR

`synchronized` + tryLock/timeout/interruptible/fairness/conditions. `lock(); try { … } finally { unlock(); }` always.

12.9 Try-lock & timed locking

What it is

`tryLock()` takes the lock only if free *now*; `tryLock(timeout, unit)` waits up to the timeout (interruptibly), returning `false` on failure.

The problem it solves

Plain `lock()` waits forever (hung requests, permanent deadlocks). Try-lock lets a thread **give up** — fail fast, retry, return "busy" — and breaks deadlock's *hold-and-wait* condition (12.17).

How it works

`true` → `try { work } finally { unlock(); }`; `false` → fallback. With several locks: if any fails, **release all held**, back off **randomly**, retry.

Java example

In `locks/Main.java`, `printer.tryLock(100, TimeUnit.MILLISECONDS)` fails while `main` holds it and prints `worker: printer busy for 100 ms, doing something else instead`. `transferTryLock` in 12.17 uses try-lock + random back-off.

Trade-offs

Pros	Cons
No infinite waits; breaks deadlock; predictable latency	Caller handles failure; retries without random back-off can **livelock**; timeout choice is a judgement call

When not to use it

When failing is unacceptable and waiting is fine (background batch job) — plain `lock()`.

Common mistakes

`unlock()` after a `false` → `IllegalMonitorStateException`; tight retry loops; expecting untimed `tryLock()` to be fair — it *barges*; `tryLock(0, unit)` respects fairness.

Real-world examples

Distributed locks (Redis `SET key value NX PX 30000`, Redisson `tryLock(wait, lease, unit)`, ZooKeeper); ticket seat holds that fail fast.

In the interview

> **30-second answer:** "`tryLock` with a timeout lets a thread give up instead of waiting forever. I use it when latency matters or when I need several locks: if I can't get all of them, I release what I hold, back off randomly and retry — which prevents deadlock."

Self-check

- **Q:** `tryLock(0, SECONDS)`? **A:** Tries once without waiting, but respects a fair queue.

TL;DR

Acquire only if available (within a timeout). Handle `false`, release everything on failure, back off randomly.

12.10 Read-write locks: `ReadWriteLock` and `StampedLock`

What it is

A shared **read lock** (many readers) plus an exclusive **write lock** (one writer, no readers): `ReentrantReadWriteLock`, and since Java 8 `StampedLock` with cheap *optimistic reads*.

The problem it solves

A mutex makes readers block each other; for **read-heavy** data (config, caches, catalogues) reads can run in parallel.

How it works

Readers share `readLock()` when no writer holds it; a writer waits for **no readers and no writer**. `ReentrantReadWriteLock` allows **downgrade** (write → read) but **not upgrade** (read → write deadlocks on itself). `StampedLock`: `tryOptimisticRead()` → copy fields to locals → `validate(stamp)`; on failure retry with `readLock()`.

Java example

`ConfigStore` and `Point` (optimistic read) in `locks/Main.java`; three readers inside the read lock at once print `3 readers held the read lock at the same time: true`.

Trade-offs

	Mutex	ReentrantReadWriteLock	StampedLock
Parallel readers	No	Yes	Yes (optimistic reads take no lock)
Reentrant	Yes	Yes	**No**
Conditions	Yes	Write lock only	No
Overhead	Lowest	Higher bookkeeping	Low for optimistic reads
Best when	Short or write-heavy	Reads ≫ writes, non-trivial reads	Very read-heavy, tiny sections

When not to use it

Write-heavy or tiny sections (bookkeeping makes it slower); when a **copy-on-write snapshot** is simpler (immutable map in a `volatile` field, replaced on write — 12.23).

Common mistakes

Upgrading read → write; writing under the read lock; using optimistic data before `validate` or without copying to locals.

Real-world examples

Caches, config holders, routing tables; `ConcurrentHashMap` goes further — lock-free reads.

In the interview

> **30-second answer:** "A read-write lock lets many readers in together but gives writers exclusive access. It helps when reads vastly outnumber writes. `ReentrantReadWriteLock` allows downgrade but not upgrade; `StampedLock` adds optimistic reads that take no lock and are validated afterwards, but it's not reentrant."

Self-check

- **Q:** Two readers and a writer arrive — who runs together? **A:** The two readers; the writer alone.

TL;DR

Many readers OR one writer, for read-heavy data. No upgrade; `StampedLock` = optimistic, non-reentrant.

12.11 Coarse-grained vs fine-grained locking

What it is

Coarse: one lock for a lot of state (the whole parking lot). **Fine**: many locks each guarding a small piece (per floor, spot, hash bucket).

The problem it solves

Balances **simplicity and safety** against **throughput under contention**.

How it works

Start coarse and measure; if contended, split **independent** state (**lock splitting**); for big collections use **lock striping** (lock `hash(i) % N`; Java 7 `ConcurrentHashMap` had 16 segments); take multiple fine locks in a **global order**; do slow work outside locks.

	Coarse	Fine
Simplicity	High	Lower
Concurrency	Low	High
Deadlock risk	Low	Higher — needs ordering
Overhead	One lock	Many locks
Cross-part invariants	Easy	Hard

Java example

Coarse: `SynchronizedCounter` (12.5). Fine: `Show` in 12.25 (one lock per seat, sorted order); `StripedHashMap` in Chapter 13.

Trade-offs

Correct first; go fine-grained only where profiling shows contention.

When not to use it

Don't split when an invariant spans the parts ("total booked ≤ capacity") unless you handle the cross-part check.

Common mistakes

Premature fine-graining; inconsistent lock order; fine locks that still funnel through one shared counter (use `LongAdder`).

Real-world examples

Java 8 `ConcurrentHashMap` locks one bin head; DB row vs table locks.

In the interview

> **30-second answer:** "I start with one lock for correctness, then refine where contention matters: per-floor or per-spot locks for a parking lot, per-show or per-seat locks for booking, striping for maps. Finer locks raise throughput but need a consistent lock order to avoid deadlock."

Self-check

- **Q:** Lock striping? **A:** N locks; each item maps to one by hash — fine-grained with a bounded lock count.

TL;DR

Coarse = simple, low throughput; fine = parallel, complex, deadlock-prone. Start coarse, measure, order locks.

Part D — Coordination

12.12 Semaphores

What it is

A **semaphore** holds **permits**: `acquire()` takes one (waiting if none), `release()` returns one. One permit ≈ a mutex, but with **no owner** — any thread may release.

The problem it solves

Limiting concurrency (≤ N DB connections, API calls) and **signaling** (0 permits = "wait until someone signals", used heavily in Chapter 13).

How it works

`new Semaphore(n)` is **non-fair** (callers may barge); `new Semaphore(n, true)` is FIFO (untimed `tryAcquire()` still barges; `tryAcquire(0, SECONDS)` doesn't). Also `tryAcquire(timeout)`, `acquire(k)`. Always `release()` in `finally` after a successful `acquire()`.

Java example

```
// File: code/lld/ch12/coordination/Main.java
        Semaphore connections = new Semaphore(3);
        // ... 10 tasks on a pool of 10 threads, each doing:
                try {
                    connections.acquire();                          // blocks when 0 permits left
                    try {
                        int now = inUse.incrementAndGet();
                        maxSeen.accumulateAndGet(now, Math::max);
                        Thread.sleep(30);                           // "run a query"
                        inUse.decrementAndGet();
                    } finally {
                        connections.release();                      // give the permit back
                    }
                } catch (InterruptedException e) { Thread.currentThread().interrupt(); }
```

Output: `10 queries done, never more than 3 at once: true`. The same file demos wait/notify, `Condition`, latch and barrier (12.13–12.14). Full runnable code: code/lld/ch12/coordination/ (see the Full Edition).

Trade-offs

Pros	Cons
Simple concurrency cap; doubles as a signal	No owner — buggy extra releases grow permits; protects no data by itself

When not to use it

As a mutex around shared data when a lock would do (locks have ownership and reentrancy).

Common mistakes

Releasing when `acquire` never succeeded (e.g. interrupted) → permit leak upward; not releasing on exceptions → permits vanish, system stalls.

Real-world examples

Bulkheads (Resilience4j `SemaphoreBulkhead`, Hystrix semaphore isolation); HikariCP's max pool size.

In the interview

> **30-second answer:** "A semaphore is a permit counter: acquire takes a permit or waits, release returns one. I use it to cap concurrent access to a limited resource — a bulkhead — or with zero initial permits as a signal from one thread to another. Unlike a lock it has no owner."

Self-check

- **Q:** `new Semaphore(0).acquire()`? **A:** Waits until another thread calls `release()` — a signal.

TL;DR

N permits: cap concurrency or signal. No ownership — release carefully in `finally`.

12.13 Condition variables: `wait/notify` and `Condition`

What it is

A **condition variable** lets a thread *sleep until a condition on shared state is true*, releasing the lock meanwhile, and lets another thread *wake it* after changing that state. Built-in: `wait()` / `notify()` / `notifyAll()` inside `synchronized`; explicit: `lock.newCondition()` with `await()` / `signal()` / `signalAll()`.

The problem it solves

Waiting for a state change without **busy-waiting** and without missing the moment.

How it works

The canonical pattern — memorize it:

```
synchronized (lock) {                 // or: lock.lock(); try {
    while (!conditionHolds()) {       // ALWAYS a while loop, never if
        lock.wait();                  //   cond.await();  releases the lock while waiting
    }
    doAction();                       // condition is true AND we hold the lock
    lock.notifyAll();                 //   cond.signal()/signalAll() if we changed state others wait for
}                                     // } finally { lock.unlock(); }
```

`wait()` **atomically releases the lock** and sleeps; it **re-acquires** it before returning. Why `while`? **Spurious wakeups** are allowed; another thread may grab the state first; with `notifyAll` only one waiter can win. `notify()` wakes one arbitrary waiter — with producers *and* consumers on one monitor it can wake the wrong kind and everyone sleeps. Use `notifyAll()`, or separate `Condition`s (`notFull`, `notEmpty`) with targeted `signal()`.

Java example

`Mailbox` (monitor, `while (item != null) wait();`) and `TurnTable` (Lock + `Condition`) in `coordination/Main.java`. Output: `received in order: [m1, m2, m3]` and `PING PONG PING PONG PING PONG`.

Trade-offs

	`wait/notify`	`Lock` + `Condition`
Conditions per lock	One wait set	Many
Targeted wake-up	No — use `notifyAll`	Yes
Timed wait	`wait(ms)`	`await(t, unit)`, `awaitNanos`
Verbosity	Less	More

When not to use it

Usually `BlockingQueue`, `Semaphore`, `CountDownLatch` or `CompletableFuture` already solve it; hand-roll only for custom coordination.

Common mistakes

`if` instead of `while`; `wait`/`notify` without holding that monitor → `IllegalMonitorStateException`; changing state without notifying (**lost wakeup**); waiting on one object while synchronized on another.

Real-world examples

`ArrayBlockingQueue`: one `ReentrantLock`, two `Condition`s (`notEmpty`, `notFull`) — built by hand in Chapter 13.

In the interview

> **30-second answer:** "A condition variable lets a thread release a lock and sleep until another thread changes the state and signals it. I always wait in a while loop that re-checks the condition, because of spurious wakeups and competing threads. With one monitor I use notifyAll; with ReentrantLock I use separate Conditions and targeted signal."

Self-check

- **Q:** What does `wait()` do with the monitor? **A:** Releases it while sleeping; re-acquires before returning.

TL;DR

Release lock, sleep until signalled, re-acquire, re-check. Always `while`; `notifyAll` or separate `Condition`s.

12.14 `CountDownLatch` and `CyclicBarrier`

What it is

`CountDownLatch(n)`: a one-time gate — `await()` returns after *n* `countDown()` calls; can't be reset. **`CyclicBarrier(n, action)`**: a reusable meeting point — when the *n*th thread calls `await()`, the action runs once, all are released, and it resets.

The problem it solves

Coordinating phases: start after all services are up (latch); finish step 1 before step 2 (barrier); release all test threads at once (latch of 1 — used throughout this book to maximise contention).

How it works

	`CountDownLatch`	`CyclicBarrier`
Who waits	Anyone calling `await()`	The *n* parties themselves
Who counts	Anyone calling `countDown()`	Arrival at `await()`
Reusable	No	Yes
Extra	Reaching 0 is permanent	Action once per trip; a failed party **breaks** it (`BrokenBarrierException`; `reset()` repairs)
Happens-before	Before `countDown()` HB after `await()`	Before `await()` HB the action HB after release

`Phaser` (Java 7) generalises both with dynamic parties across many phases.

Java example

`latchDemo` / `barrierDemo` in `coordination/Main.java` print `all services ready: true (count now 0)` and `phase 1 complete…`, `phase 2 complete…`.

Trade-offs

Latch: simple, one-shot. Barrier: reusable but fragile — one failure breaks it for all.

When not to use it

To wait for N tasks and collect results, `invokeAll` or `CompletableFuture.allOf` is clearer.

Common mistakes

A worker throws before `countDown()` → waiting forever (put it in `finally`; use `await(timeout)`); fewer threads than barrier parties.

Real-world examples

Test harnesses releasing N threads together; start-up waiting for caches, DB pools and consumers.

In the interview

> **30-second answer:** "`CountDownLatch` is a one-shot gate that opens after N countDowns; `CyclicBarrier` is a reusable rendezvous where N threads wait for each other each phase, optionally running an action when all arrive."

Self-check

- **Q:** 3 workers syncing at the end of every simulation step? **A:** `CyclicBarrier(3)`.

TL;DR

Latch = one-shot "wait for N events"; barrier = reusable "wait for N threads". `countDown` in `finally`; use timeouts.

Part E — Lock-free

12.15 Compare-and-swap (CAS) & atomics

What it is

CAS is one CPU instruction (`LOCK CMPXCHG` on x86; LL/SC on ARM): "if the value is still `expected`, set it to `new` and succeed; else fail." Java exposes it via `AtomicInteger`, `AtomicLong`, `AtomicBoolean`, `AtomicReference`, `AtomicIntegerArray`, `LongAdder` and `VarHandle`.

The problem it solves

Atomic updates **without locks**: no blocking, no deadlock, usually faster under low–moderate contention; the basis of the JDK's concurrency utilities.

How it works

Retry loop: read → compute → `compareAndSet(current, next)` → on failure retry. Helpers: `incrementAndGet`, `updateAndGet(fn)`, `accumulateAndGet(x, fn)` (the function may run several times — keep it side-effect-free). `LongAdder` spreads contended updates over cells and sums them (fast hot counters; `sum()` isn't an atomic snapshot). **ABA problem:** A → B → A makes a stale CAS succeed; fix with a version — `AtomicStampedReference` / `AtomicMarkableReference` (GC prevents classic node-reuse ABA for fresh nodes in Java).

Java example

```java
// File: code/lld/ch12/atomics/Main.java
    static int casIncrement(AtomicInteger a) {
        while (true) {
            int current = a.get();                  // 1. read
            int next = current + 1;                 // 2. compute
            if (a.compareAndSet(current, next)) {   // 3. write only if nobody changed it meanwhile
                return next;
            }                                       // 4. someone won the race: retry
        }
    }
```

The file also has a lock-free **Treiber stack** (push/pop swing one `AtomicReference` head with CAS), `LongAdder`, and an ABA demo.

Output: counter `400000`, stack popped `40000`; after A→B→A a plain CAS **succeeds** (`true`) while the stamped CAS **fails** (`false`).

Full runnable code: code/lld/ch12/atomics/ (see the Full Edition).

Trade-offs

	Locks	CAS / atomics
Blocking	Yes	No (retry)
Deadlock	Possible	Impossible
Scope	Any number of variables	One variable per CAS
Heavy contention	Threads sleep	Wasted retries — use `LongAdder`/striping
Difficulty	Moderate	Hard beyond counters/flags/one reference

When not to use it

Multi-variable invariants (moving money between two accounts) — use a lock, or put all fields in one immutable object and CAS its reference (Chapter 13's token bucket).

Common mistakes

`get()` then `set()` (still a race); side effects in `updateAndGet` lambdas; `AtomicLong` for very hot metrics (use `LongAdder`).

Real-world examples

`ConcurrentHashMap` CAS into empty bins; `ConcurrentLinkedQueue` (Michael–Scott); AQS state; JPA `@Version` / `UPDATE ... WHERE version = ?` (CAS in the DB); LMAX Disruptor.

In the interview

> **30-second answer:** "CAS atomically updates a value only if it still equals what I read; otherwise I retry. Java's atomic classes wrap it, so counters, flags and single references can be updated without locks and without deadlock. It covers one variable at a time, can suffer ABA — fixed with a stamped reference — and under heavy contention I prefer `LongAdder`."

Self-check

- **Q:** A failed `compareAndSet` means? **A:** Someone changed the value since you read it; re-read and retry.

TL;DR

CAS = update only if unchanged, retried. Lock-free, deadlock-free, one variable. Watch ABA; `LongAdder` for hot counters.

12.16 Concurrent collections

What it is

Need	Use	Key behaviour
Map	`ConcurrentHashMap`	Lock-free reads; writes lock one bin; atomic `putIfAbsent`, `computeIfAbsent`, `compute`, `merge`; **no nulls**; weakly consistent iterators
Sorted map/set	`ConcurrentSkipListMap` / `Set`	Lock-free, sorted, O(log n)
List read ≫ write	`CopyOnWriteArrayList` / `Set`	Writes copy the array; iteration uses a snapshot
Non-blocking queue	`ConcurrentLinkedQueue` / `Deque`	Lock-free; `poll()` returns null when empty
Blocking queue	`ArrayBlockingQueue`, `LinkedBlockingQueue`, `PriorityBlockingQueue`, `DelayQueue`, `SynchronousQueue`, `LinkedTransferQueue`	`put` waits when full, `take` when empty

Two `ConcurrentHashMap` guarantees to quote: **`computeIfAbsent`/`compute`/`merge` are atomic** — the mapping function runs **exactly once if absent, else not at all** (other updates to that bin block meanwhile, so keep it short and don't touch the same map; `ConcurrentSkipListMap`'s default may call it more than once). **Iterators are weakly consistent** — never throw CME, may or may not show later updates; `size()` is an estimate.

The problem it solves

One lock around a `HashMap` serialises everything; concurrent collections allow parallel access plus **atomic compound operations**.

How it works

Pick by access pattern; replace check-then-act with one call: `putIfAbsent`, `merge(k, 1, Integer::sum)`, `computeIfAbsent(k, ...)`; hand off work with a `BlockingQueue` (12.22). Sequences of calls are still not atomic.

Java example

`collections/Main.java` counts words with `merge(w, 1, Integer::sum)` from 4 threads, creates per-user lists with `computeIfAbsent`, and claims a seat with `putIfAbsent`:

```
word counts (4 threads x 1000 reps): {apple=12000, banana=8000, cherry=4000}
first putIfAbsent returns : null
second putIfAbsent returns: bob  (seat already bob's)
```

Full runnable code: code/lld/ch12/collections/ (see the Full Edition).

Trade-offs

Collection	Great for	Bad for
`ConcurrentHashMap`	Shared maps, caches, counters	Multi-key atomicity; `size()` estimate
`CopyOnWriteArrayList`	Listeners, rarely changed config	Frequent writes (O(n) copy)
`Collections.synchronizedMap`	Legacy, low contention	High contention; manual iteration locking
`ArrayBlockingQueue`	Bounded hand-off, back-pressure	Very high throughput (one lock)
`LinkedBlockingQueue`	Higher throughput (two locks)	Default is **unbounded**

When not to use it

Thread-confined objects (plain collections are faster); multi-key transactions (one lock or a DB).

Common mistakes

Slow work or same-map calls inside `computeIfAbsent` (bin locked; can throw `IllegalStateException` or deadlock); `null` in a `ConcurrentHashMap` → NPE.

Real-world examples

Caffeine on `ConcurrentHashMap`; Guava `EventBus` subscribers in a `CopyOnWriteArraySet`.

In the interview

> **30-second answer:** "I use `ConcurrentHashMap` for shared maps and its atomic methods — `putIfAbsent`, `computeIfAbsent`, `merge` — instead of check-then-act. `CopyOnWriteArrayList` for read-mostly lists like listeners. `BlockingQueue`, preferably bounded, to hand work between threads."

Self-check

- **Q:** Atomic word count? **A:** `map.merge(word, 1, Integer::sum)`.

TL;DR

CHM + atomic methods; COW for read-mostly lists; bounded `BlockingQueue` for hand-offs. Single calls atomic, sequences not.

Part F — Liveness failures

12.17 Deadlock

What it is

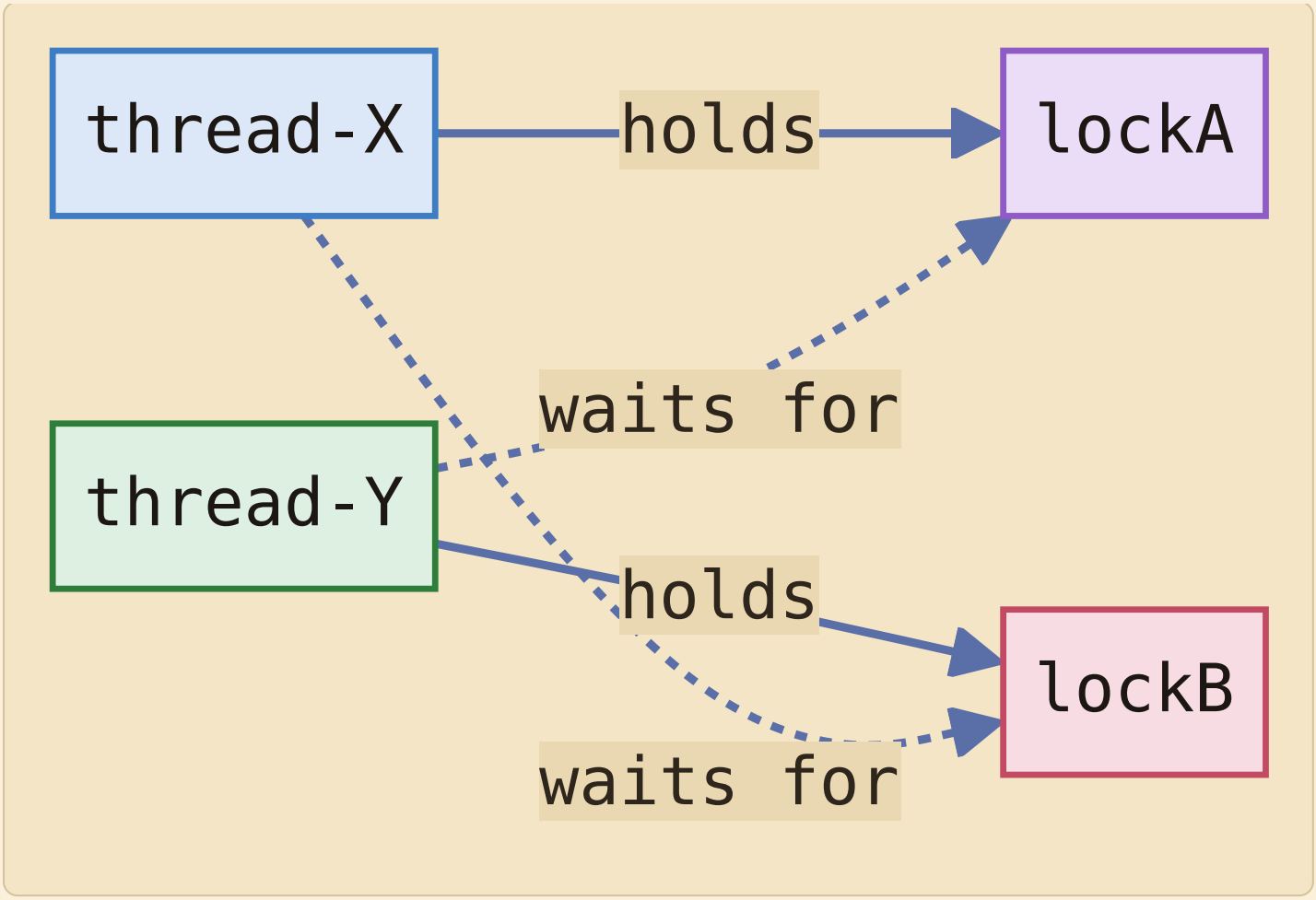
Two or more threads each wait forever for a resource another holds.

The problem it solves

Understanding it lets you *design it out*. All four **Coffman conditions** (1971) are required:

#	Condition	Meaning	How to break it
1	**Mutual exclusion**	One holder per resource	Lock-free/immutable data, shareable read locks
2	**Hold and wait**	Holding one while waiting for another	Acquire all at once, or `tryLock` and release on failure
3	**No preemption**	Only the holder releases	Timeouts, expiring leases
4	**Circular wait**	T1 waits for T2, T2 for T1	**Global lock ordering**

Break **any one** and deadlock is impossible.



How it works

1. **Lock ordering** (most common): a total order (account id, seat id, `identityHashCode` + tie-breaker lock).
2. **Try-lock**: `tryLock(timeout)`; on failure release all, random back-off, retry.
3. **Avoid nested locks**: call alien code (listeners) outside your lock.
4. **Detection**: `jstack` / `jcmd` print "Found one Java-level deadlock"; `ThreadMXBean.findDeadlockedThreads()`.
5. **Avoidance** (Banker's algorithm) — OS theory, rarely used in apps.

Java example

```
// File: code/lld/ch12/deadlock/Main.java
    static void transferOrdered(Account from, Account to, int amount) {
        Account first = from.id < to.id ? from : to;
        Account second = from.id < to.id ? to : from;
        first.lock.lock();
        try {
            second.lock.lock();
            try {
                if (from.balance >= amount) { from.balance -= amount; to.balance += amount; }
            } finally { second.lock.unlock(); }
        } finally { first.lock.unlock(); }
    }
```

`transferTryLock` instead `tryLock`s both with 1 ms timeouts, releases on failure, and sleeps a random back-off.

`demonstrateDeadlock` forces opposite-order locking with a barrier and detects it:

```
deadlock detected between: [thread-X, thread-Y]  (states BLOCKED, BLOCKED)
lock ordering: finished, total money = 2000
tryLock + back-off: finished, total money = 2000
```

Full runnable code: code/lld/ch12/deadlock/ (see the Full Edition).

Trade-offs

Strategy	Pros	Cons
Lock ordering	Simple, zero cost	Needs a stable global order
Try-lock + back-off	No order needed; bounded waits	Retry logic; livelock without randomness
Single coarse lock	No deadlock among your locks	Lower throughput
Lock-free / immutable	No locks	Harder design

When not to use it

N/A — always prevent deadlock; prefer simple lock ordering to detection/recovery in app code.

Common mistakes

Calling listeners or another object's synchronized method under your lock; ordering by `hashCode()` without tie handling; **thread-starvation deadlock** — pool tasks that submit sub-tasks to the *same* pool and wait.

Real-world examples

InnoDB/PostgreSQL detect deadlocks and abort a victim; bank transfers lock accounts in id order.

In the interview

> **30-second answer:** "Deadlock needs four conditions: mutual exclusion, hold-and-wait, no preemption, and circular wait. I break circular wait with a global lock order — e.g. lock seats or accounts in ascending id — or break hold-and-wait with tryLock with timeout and release-and-retry. I avoid calling external code while holding a lock, and I'd detect it in production with thread dumps."

Self-check

- **Q:** What does lock ordering break? **A:** Circular wait.

TL;DR

Deadlock = cycle of waiting; needs all 4 Coffman conditions. Lock ordering or tryLock + release. No alien code under a lock.

12.18 Livelock

What it is

Threads keep running and reacting to each other but make **no progress**.

The problem it solves

Recognising it — it often comes *from* deadlock avoidance: two threads `tryLock`, fail, release and retry at the same rhythm.

How it works (and how to fix it)

T1 takes A, fails on B, releases; T2 takes B, fails on A, releases; both retry → forever. **Fixes:** random or exponential back-off, a consistent lock order (no retries needed), or a tie-breaker (older transaction wins).

Java example

`transferTryLock` sleeps a **random** number of nanoseconds after failure; remove it and the threads could fail in lock-step.

Trade-offs

Random back-off makes progress very likely but adds latency; exponential back-off adapts to contention.

When not to use it

Never design retry loops with no randomness or tie-breaker.

Common mistakes

Fixed retry delays for everyone; retrying forever with no limit or alert.

Real-world examples

Ethernet CSMA/CD binary exponential back-off; AWS SDK and Resilience4j retries with **jitter**.

In the interview

> **30-second answer:** "Livelock is when threads keep reacting to each other without progress — like two people side-stepping in a corridor. It often comes from symmetric retry logic; I fix it with randomized or exponential back-off, or remove retries by using a lock order."

Self-check

- **Q:** Thread states in deadlock vs livelock? **A:** Deadlocked threads are blocked/waiting; livelocked ones are `RUNNABLE` but achieve nothing.

TL;DR

Busy but no progress. Break symmetry with jittered back-off or a tie-breaker.

12.19 Starvation (and fairness)

What it is

A thread waits indefinitely because others keep winning, though the system progresses.

The problem it solves

Recognising the causes: barging locks under contention, writers behind endless readers, low priorities, a long task hogging a single-thread executor.

How it works (fixes)

Fair locks/semaphores where it matters (`new ReentrantLock(true)`, `new Semaphore(n, true)`, `new ReentrantReadWriteLock(true)`); short critical sections; separate pools for slow and fast work (bulkheads); don't rely on `Thread.setPriority` (OS-dependent).

Java example

Chapter 13's Dining Philosophers `Waiter` uses `new Semaphore(n - 1, true)`.

Trade-offs

Fairness prevents starvation but reduces throughput.

When not to use it

Don't make every lock fair "just in case".

Common mistakes

Confusing it with deadlock: in starvation *someone* progresses; in deadlock nobody does.

Real-world examples

OS schedulers' *aging*; Linux CFS and EEVDF (kernel 6.6+) fair CPU shares.

In the interview

> **30-second answer:** "Starvation is when a thread never gets a resource because others always win — e.g. writers behind endless readers. Fair locks, short critical sections and separate pools fix it, at some throughput cost."

Self-check

- **Q:** Fix for writer starvation in `ReentrantReadWriteLock`? **A:** Fair mode (`new ReentrantReadWriteLock(true)`).

TL;DR

Some thread never gets a turn. Fairness, short sections, separate pools.

Part G — Concurrency patterns

The next five patterns share one runnable file, `patterns/Main.java`. Full runnable code: code/lld/ch12/patterns/ (see the Full Edition).

12.20 Signaling pattern

What it is

One thread **notifies** another that an event happened: "B must run after A".

The problem it solves

Ordering between threads without busy-waiting on a flag.

How it works

Create a signal in the "not yet" state; the waiter blocks on it; the signaller releases it; the waiter continues and (via happens-before) sees everything A did.

Tool	Reusable	Carries a value	Good for
`Semaphore(0)`	Yes	No	Ping-pong (FooBar, Ch 13)
`CountDownLatch(1)`	No	No	One-time "go"
`CompletableFuture`	No	Yes	"Here is the result"
Condition variable	Yes	Via shared state	Complex conditions

Java example

`signaling()`: B starts first but blocks on `aDone.acquire()` (a `Semaphore(0)`); A logs `charge payment` and calls `release()`, so A's line always prints first.

Trade-offs

Semaphores are compact but easy to mis-count; latches can't reset; futures are clean but single-use.

When not to use it

If the signal is really "here's a work item", use a `BlockingQueue`.

Common mistakes

A tool that *forgets* signals (bare `notify()` before the waiter waits = lost wakeup); semaphores and latches remember.

Real-world examples

`CompletableFuture` chains in WebFlux/Netty callbacks; start-up gates; "order paid → send confirmation".

In the interview

"Signaling means one thread waits for an event raised by another. I use a zero-permit semaphore for repeated hand-offs, a latch for one-time events, or a `CompletableFuture` when a result travels with the signal."

Self-check

Q: Why is bare `notify()` risky? **A:** If it happens before the wait, it's lost; pair it with a state variable checked in a loop.

TL;DR

"You may go now": Semaphore(0), latch, or future — make signals unlosable.

12.21 Thread pool pattern

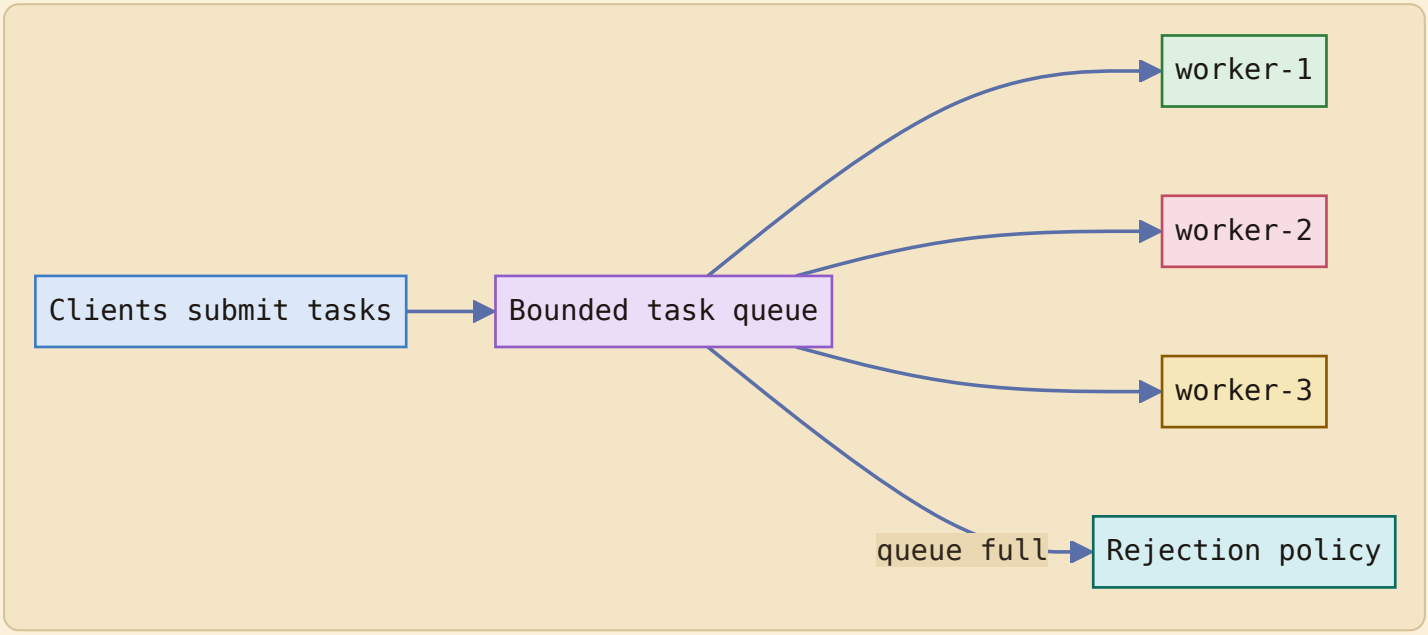
What it is

A fixed set of **worker threads** takes tasks from a shared **queue**, reusing threads.

The problem it solves

Per-task thread creation is expensive, and unbounded concurrency overwhelms CPU, memory and downstreams.

How it works



Workers loop `queue.take(); task.run()`. **`ThreadPoolExecutor` rules:** below **core** threads → new thread per task; then the **queue**; queue full → extra threads up to **max**; still full → **rejection policy** (`AbortPolicy`

default throws, `CallerRunsPolicy` runs on the submitter, `DiscardPolicy`, `DiscardOldestPolicy`). Shutdown drains the queue (one **poison pill** per worker in the hand-made pool); `submit`'s "shut down? else enqueue" is check-then-act, so it shares a lock with pill-adding or a task could land behind the pills and never run. **Sizing:** CPU-bound ≈ cores; I/O-bound ≈ cores × (1 + wait/compute), or virtual threads.

Java example

```
// File: code/lld/ch12/patterns/Main.java
        ThreadPoolExecutor exec = new ThreadPoolExecutor(
                2, 4,                                   // core and max threads
                30, TimeUnit.SECONDS,                   // idle time before extra threads die
                new ArrayBlockingQueue<>(100),          // bounded: never an OOM from a huge backlog
                new ThreadPoolExecutor.CallerRunsPolicy()); // when full, the submitter runs it (back-pressure)
```

(`SimpleThreadPool` in the same file is a hand-made pool showing the mechanics.)

Trade-offs

Choice	Effect
Bounded queue	Protects memory; needs a rejection policy
Unbounded (`newFixedThreadPool`)	Never rejects, may OOM; max size never used
`CallerRunsPolicy`	Natural back-pressure
Many pools (bulkheads)	Isolation; more threads overall

When not to use it

Huge numbers of blocking tasks on Java 21 (virtual-thread-per-task); recursive CPU work (`ForkJoinPool`, work stealing).

Common mistakes

Tasks waiting on tasks in the same bounded pool (thread-starvation deadlock); swallowed exceptions; no `shutdown()`.

Real-world examples

Tomcat `maxThreads` (default 200), Spring `ThreadPoolTaskExecutor`, DB connection pools, `ForkJoinPool.commonPool()`.

In the interview

"A thread pool reuses a bounded set of workers fed by a queue. I size it by workload type, use a bounded queue with an explicit rejection policy for back-pressure, and give slow dependencies their own pool as a bulkhead."

Self-check

Q: When are threads beyond `corePoolSize` created? **A:** Only when the queue is full (and below max).

TL;DR

Workers + queue + rejection policy. Bound everything; size by CPU vs I/O.

12.22 Producer-consumer pattern

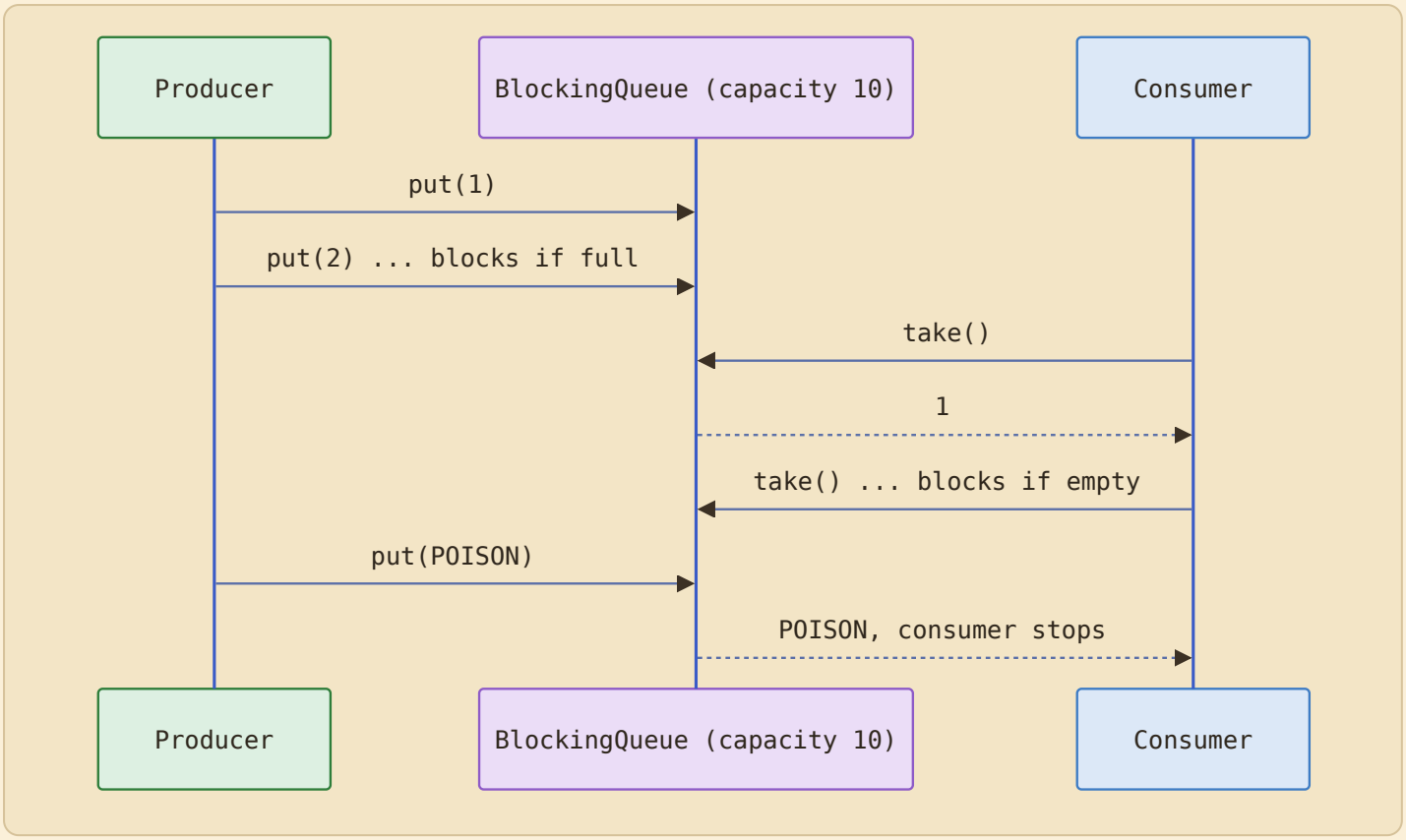
What it is

Producers put items in a shared **buffer**; **consumers** take them out; the buffer decouples their speeds.

The problem it solves

Different rates: a bounded buffer smooths bursts and applies **back-pressure**, with neither side knowing the other.

How it works



Share a bounded `BlockingQueue`; `put` waits when full, `take` when empty. Shutdown: a **poison pill per consumer**, interruption, or a done flag + `poll(timeout)`.

Java example

`producerConsumer()`: 1 producer, 2 consumers, `ArrayBlockingQueue<>(10)`, one `POISON` per consumer; the sum is exactly `500500`. Chapter 13 builds the queue itself two ways.

Trade-offs

Pros	Cons
Decouples rates; back-pressure; easy to scale consumers	Queueing latency; no ordering across consumers; needs a shutdown protocol

When not to use it

When the caller needs the result immediately — call directly.

Common mistakes

Unbounded buffer (memory blow-up); one poison pill for N consumers (N−1 hang); busy-polling `poll()` instead of `take()`.

Real-world examples

Kafka across machines (topics = buffer; consumer groups scale), and inside its Java producer (accumulator + sender thread).

In the interview

"Producer-consumer decouples producers from consumers with a bounded blocking queue: put waits when full, take waits when empty, giving natural back-pressure. I stop consumers with one poison pill each, and scale consumers independently."

Self-check

Q: Why bounded? **A:** So a slow consumer pushes back instead of the queue growing until memory runs out.

TL;DR

Bounded `BlockingQueue` between them; back-pressure for free; one poison pill per consumer.

12.23 Reader-writer pattern

What it is

Many readers may access shared data together; a writer needs exclusive access.

The problem it solves

Read-mostly data (config, catalogues, routing tables) shouldn't be serialised through one mutex.

How it works

(1) **Read-write lock** (12.10), fair if writers could starve. (2) **Copy-on-write snapshot**: an immutable object in a `volatile` field; readers just read it; a writer builds a new copy and swaps the reference.

```
private volatile Map<String, String> config = Map.of();       // immutable snapshot
String get(String k) { return config.get(k); }                  // readers: no lock
synchronized void put(String k, String v) {                      // writers: serialized
    Map<String, String> copy = new HashMap<>(config);
    copy.put(k, v);
    config = Map.copyOf(copy);                                    // publish new snapshot
}
```

	RW lock	Copy-on-write snapshot
Reader cost	Shared lock	One volatile read
Writer cost	Exclusive lock, in-place	Copy whole structure
Readers see	Latest data	Consistent, maybe slightly old snapshot
Best for	Frequent-ish writes, big data	Rare writes, small–medium data

Java example

`ConfigStore` (RW lock) in `locks/Main.java`; the snapshot above; `CopyOnWriteArrayList` in `collections/Main.java`.

Trade-offs

See the table; RW-lock readers must not upgrade.

When not to use it

Write-heavy data — a mutex or `ConcurrentHashMap`.

Common mistakes

Mutating the shared snapshot in place; read → write upgrades.

Real-world examples

Feature-flag/config clients swapping immutable snapshots; Linux RCU (read-copy-update).

In the interview

"For read-heavy data I use either a read-write lock or, if writes are rare, an immutable snapshot in a volatile field that writers replace — readers never block."

Self-check

Q: What guarantees readers see a fully built snapshot? **A:** The volatile write HB later volatile reads, and the snapshot is immutable.

TL;DR

Many readers, one writer: RW lock, or immutable snapshot + volatile swap.

12.24 Immutability & thread confinement

What it is

Thread safety **without synchronization**. **Immutability**: state can't change after construction (`String`, `LocalDate`, `BigDecimal`, records of immutables, `List.of`). **Thread confinement**: only one thread touches the data (locals, `ThreadLocal`, a single owner thread such as an event loop or actor).

The problem it solves

Races need *shared mutable* state — remove "mutable" or remove "shared".

How it works

Immutable checklist: `final` class or `record`; `private final` fields; no setters (return new objects, `withItem`); defensive-copy mutable inputs (`List.copyOf`) and never expose internals; don't leak `this`. **Confinement:** locals; `ThreadLocal.withInitial(...)` (call `remove()` in pools); a single-writer thread fed by a queue.

Java example

```java
// File: code/lld/ch12/patterns/Main.java
    record Order(String id, List<String> items) {
        Order { items = List.copyOf(items); }                 // caller's list can't change ours
        Order withItem(String item) {                          // "modify" = return a new object
            List<String> copy = new ArrayList<>(items); copy.add(item);
            return new Order(id, copy);
        }
    }

    // ---------- 5. Thread confinement ----------
    private static final ThreadLocal<StringBuilder> BUFFER = ThreadLocal.withInitial(StringBuilder::new);
```

The caller's later `mutable.add("hacked")` doesn't affect `o1` (output `o1 = [pen], o2 = [pen, book]`).

Trade-offs

	Immutability	Confinement
Cost	New object per change (GC)	Per-thread copies; `ThreadLocal` leaks if not cleaned
Sharing	Share freely	Not shareable
Reasoning	Easiest	Easy, if truly enforced

When not to use it

Very large objects changed very often — use a concurrent structure.

Common mistakes

`final List` that is mutable and exposed; `ThreadLocal` in pools without `remove()` (and heavy `ThreadLocal`s with many virtual threads — **scoped values** are the alternative, preview in 21, final in Java 25 via JEP 506); sharing `SimpleDateFormat` instead of `DateTimeFormatter`.

Real-world examples

`java.time` and `DateTimeFormatter`; Netty/Vert.x event-loop confinement; Redis's single main thread; Spring `@Transactional` state in `ThreadLocal`s.

In the interview

"The easiest thread-safety is none needed: make value objects immutable — final fields, defensive copies, 'with' methods — and confine mutable state to one thread with locals, ThreadLocal, or a single owner thread fed by a queue."

Self-check

Q: Is a class with `private final Date created` immutable? **A:** Not unless it copies the mutable `Date` in and out — prefer `Instant`.

TL;DR

No shared mutable state → no races. Immutable objects share freely; confined ones aren't shared.

Part H — Concurrency in LLD interviews

12.25 How concurrency shows up in LLD interviews (parking, seat booking, inventory)

What it is

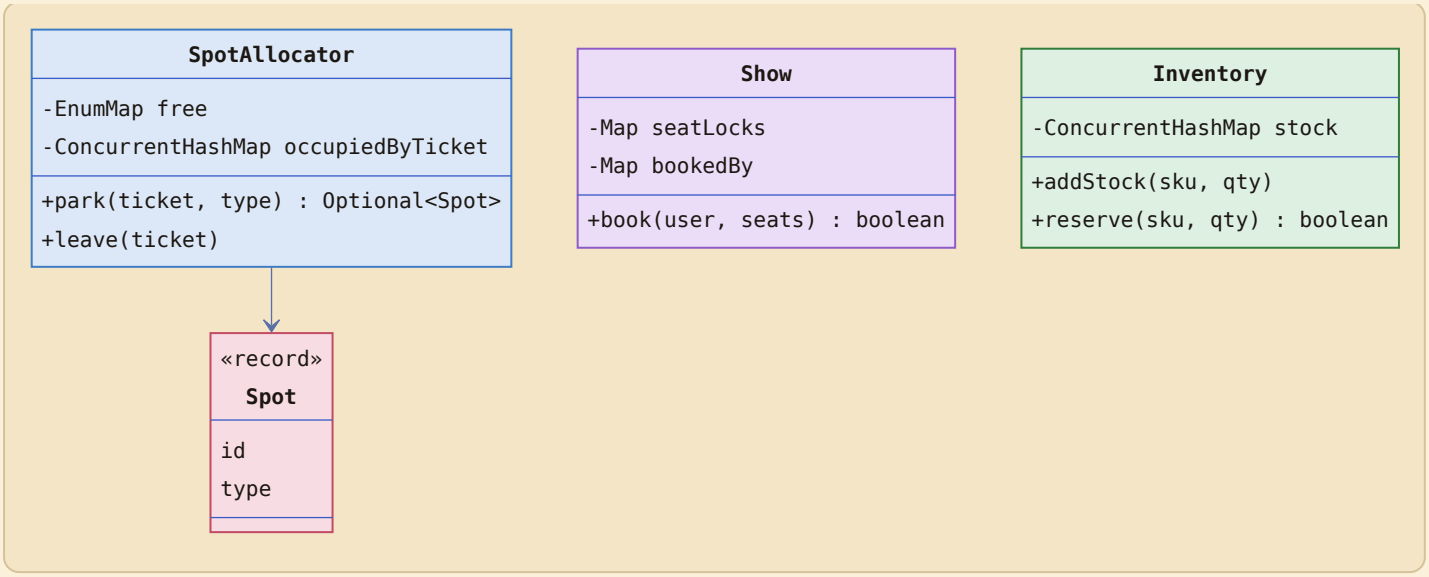
Most case studies have a **contended resource** — parking spot, seat, room, limited stock, wallet balance. The interviewer checks you prevent double allocation and still perform.

The problem it solves

Turning a single-threaded class diagram into a **thread-safe design** with the smallest correct synchronization.

How it works — a 6-step recipe

1. **Name the shared mutable state** (free spots, seat → owner, stock).
2. **State the invariant** ("a seat has at most one booking", "stock ≥ 0").
3. **Find the critical operation** (usually check-then-act: *find free → assign*).
4. **Pick the smallest atomic tool:** one item → `queue.poll()`, `putIfAbsent`, `compareAndSet`, CAS loop; several items all-or-nothing → per-item locks **in sorted order** (or a per-show/per-floor lock); across servers → DB constraints/transactions, optimistic versions, distributed locks.
5. **Choose granularity:** per floor/show/SKU, not global.
6. **Handle time:** holds with expiry (seat held 10 min during payment), timeouts, idempotent retries.



Java example

```java
// File: code/lld/ch12/lldcases/Main.java
        /** Locks seats in sorted order (prevents deadlock), checks all, then books all. */
        boolean book(String user, Set<String> seats) {
            List<ReentrantLock> held = new ArrayList<>();
            try {
                for (String id : new TreeSet<>(seats)) {         // global order: A1 < A2 < A3
                    ReentrantLock l = seatLocks.get(id);
                    if (l == null) return false;                  // unknown seat
                    l.lock();
                    held.add(l);
                }
                for (String id : seats) if (bookedBy.containsKey(id)) return false;   // someone got one first
                for (String id : seats) bookedBy.put(id, user);
                return true;
            } finally {
                for (int i = held.size() - 1; i >= 0; i--) held.get(i).unlock();
            }
        }
```

Parking keeps free spots in a `ConcurrentLinkedQueue` per type — `poll()` is an atomic claim, and `leave` uses `ConcurrentHashMap.remove` so a double exit is harmless. **Inventory** uses an `AtomicInteger` per SKU with a CAS loop that decrements only if `cur >= qty`. `raceAll` releases all tasks via a `CountDownLatch`. Output:

```
parking: 100 cars raced for 10 spots -> parked 10, distinct spots 10, free left 0
booking: 50 users wanted seat A2 -> successful bookings 1
booking: alice {A3,A4} vs bob {A4,A3} (opposite order) -> winners 1, A3 and A4 same owner: true
inventory: 200 buyers, 50 in stock -> sold 50, left 0
```

Full runnable code: code/lld/ch12/lldcases/ (see the Full Edition).

Trade-offs

Resource	In-memory	Across many servers
Parking spot	Lock-free queue per type/floor, or per-floor lock	`UPDATE ... WHERE status='FREE'` + check affected rows
Seat	Per-show/per-seat locks, sorted; hold with TTL	Unique constraint (show, seat); Redis holds (`SET NX PX`)
Inventory	`AtomicInteger` CAS / `compute`	`UPDATE stock SET qty = qty - 1 WHERE sku = ? AND qty >= 1`; expiring reservations

When not to use it

Don't bolt `synchronized` onto every method — protect only operations on contended state.

Common mistakes

A global `synchronized park()` (serialises all gates); availability check and booking as separate calls; locking seats in click order (deadlock).

Real-world examples

BookMyShow/Ticketmaster seat holds that expire (a lock with a lease); flash-sale stock decremented atomically (Redis `DECR`/Lua, conditional SQL).

In the interview

> **30-second answer:** "The critical section is 'find a free X and assign it'. Inside one JVM I make that a single atomic step — a lock-free queue poll for spots, per-seat locks taken in sorted order for multi-seat booking, a CAS decrement for stock. Across servers I'd push it into the database with a conditional update or unique constraint, and add holds with expiry for seats during payment."

Self-check

- **Q:** Which Coffman condition does sorting seat ids break? **A:** Circular wait.

TL;DR

Find the contended resource and invariant; make "check + claim" one atomic step with the smallest tool; lock fine-grained and in order; use DB/distributed mechanisms across servers.

12.26 Choosing the right tool (cheat sheet)

I need to…	Use
Increment a shared counter	`AtomicInteger` / `AtomicLong`; `LongAdder` if very hot
Stop a loop from another thread	`volatile boolean` or `interrupt()`
Make a multi-step update atomic	`synchronized` or `ReentrantLock`
Give up if the lock is busy	`tryLock(timeout)`
Many readers, few writers	RW lock, `StampedLock`, or volatile immutable snapshot
Limit concurrent access to N	`Semaphore(N)`
Wait for a condition	`Lock` + `Condition` (or `wait/notifyAll`) in a `while`
Wait for N events once	`CountDownLatch`
N threads meet each phase	`CyclicBarrier` / `Phaser`
Hand work between threads	Bounded `BlockingQueue`
Shared map	`ConcurrentHashMap` + `putIfAbsent` / `compute` / `merge`
Read-mostly list	`CopyOnWriteArrayList`
Run tasks	Bounded `ExecutorService`; virtual threads for blocking I/O
Split CPU work	`ForkJoinPool` / parallel streams
Combine async results	`CompletableFuture`
Avoid synchronization	Immutability, thread confinement

12.27 Chapter self-check

1. **Q:** Six `Thread.State` values? **A:** NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, TERMINATED.
2. **Q:** Why isn't `volatile` enough for `count++`? **A:** Visibility, not atomicity.
3. **Q:** Why `while` around `wait()`? **A:** Spurious wakeups and competing threads.
4. **Q:** Four Coffman conditions? **A:** Mutual exclusion, hold and wait, no preemption, circular wait.
5. **Q:** Deadlock vs livelock vs starvation? **A:** All wait / all run but none progress / some progress, one never does.
6. **Q:** Lock-free spot allocation across gates? **A:** Free spots in a concurrent queue; `poll()` atomically.

12.28 Chapter TL;DR

- Races come from **shared mutable state**: make critical sections atomic, or remove sharing/mutability.
- Visibility needs **happens-before**: locks, `volatile`, start/join, concurrent utilities.
- Toolbox: `synchronized` → `ReentrantLock` → RW locks → semaphores → latches/barriers → atomics/CAS → concurrent collections.
- Liveness: **lock ordering** or **tryLock + back-off** against deadlock; jitter against livelock; fairness against starvation.
- Patterns: signaling, thread pool, producer-consumer, reader-writer, immutability/confinement.
- Interviews: find the contended resource, keep "check + claim" atomic, lock fine-grained and in order, explain the multi-server version.